Split Complex Lorentzian Dynamics

Part XXV: The Lifted-Frame Torsor, Equivariant Descent, and the One-Fibre Boundary

Closing the last one-fibre action gap while isolating, rather than assuming, the missing geometry of a spin structure

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Abstract

The previous stages of the reconstruction had reached a striking but still incomplete position. The internal core had been identified group-theoretically with the unit quaternions and with $SU(2)$, the visible group with $SO(3)$, the intrinsic projection with the quaternionic rotation map, and its kernel with the two signs $\{\pm 1\}$. The cover had then been decomposed into local sections and locally constant transition signs, reconstructed globally from those data, and finally identified as a two-sheeted covering map. Part XXIV went one level higher: it reconstructed the carrier $\mathcal{F}^+(E)$ of positively oriented orthonormal frames of the inherited three-dimensional metric carrier, proved that G_{vis} acts freely and transitively on it, and constructed a two-valued lifted-frame carrier $\widetilde{\mathcal{F}}_{F_0}$ over a temporary reference frame. Yet one precise gap remained. The full internal group $\mathcal{L}$ had never been made to act on the lifted-frame carrier itself. Consequently the project had a lifted carrier and a two-sheeted fibre description, but not yet the formal one-fibre torsor diagram that standard spin geometry would suggest.

Task 25 closes exactly that gap and deliberately nothing broader. It defines directly

$$a \cdot (u, F) = (au,\, p(a) \cdot F)$$

on $\widetilde{\mathcal{F}}_{F_0}$ and proves that this is a genuine action of $\mathcal{L}$. The action is free and transitive; explicitly, the unique element carrying x to y is $y_{\mathrm{elt}} x_{\mathrm{elt}}^{-1}$. The projection $\pi : \widetilde{\mathcal{F}}_{F_0} \to \mathcal{F}^+(E)$ is equivariant along $p : \mathcal{L} \to G_{\mathrm{vis}}$, and restricting the full internal action to $K_\pm = \ker p$ gives exactly the previously reconstructed vertical sign action. Thus kernel signs act trivially downstairs but freely upstairs. The inherited carrier equivalence $\widetilde{\mathcal{F}}_{F_0} \simeq \mathcal{L}$ is also proved equivariant, so the torsor structure is attached to the actual intrinsic action rather than inferred merely from a bijection.

Reference change is then closed at the same level of precision. Between two ordinary frames F_0, F_1 there is exactly one visible transformation. Its fibre under p contains exactly two internal lifts, and those lifts form a free transitive $K_\pm$-orbit. Each chosen lift induces an equivariant transfer $\widetilde{\mathcal{F}}_{F_0} \to \widetilde{\mathcal{F}}_{F_1}$; any two such transfers differ pointwise by one and only one kernel sign. The project therefore obtains a sharp two-level statement: reference change is unique downstairs and exactly two-valued upstairs. This ambiguity is classified, not normalized away.

These results are frozen in a record `OneFiberLiftInterface` containing the ordinary frame carrier, the lifted carrier, both free/transitive actions, the equivariant projection, and the exact kernel description of its fibres, while keeping temporary reference choices outside the interface. Task 25 thereby closes the entire algebraic one-fibre problem. The formal run contains five new modules and 978 lines, exposes 34 audited endpoints, reports a focused green build of 8218 jobs and a whole-project green build of 8289 jobs, and contains no `sorry`, `admit`, custom axiom, or prohibited proof escape [2].

The comparison with established mathematics is now unusually sharp. Pointwise, the project has reconstructed the same group-theoretic pattern that underlies three-dimensional spin geometry: a visible oriented-frame torsor for a group externally identified with $SO(3)$, a lifted torsor for a group externally identified with $Spin(3) \cong SU(2)$, an equivariant two-to-one projection, and a central $\mathbb{Z}_2$ kernel. Earlier work additionally reconstructed the normalized extended group as $U(2)$, which is classically $Spin^c(3)$ at the group level. But none of these pointwise identifications is a spin structure. A spin structure is a global lift of an oriented orthonormal frame *bundle* over a base, with topology/smoothness, local triviality, and compatible group actions. Task 25 has only one completely reconstructed fibre and explicitly introduces no nontrivial base, no varying vector-space family, no total frame bundle, no principal bundle, and no characteristic class. The remaining difficulty has therefore become more concentrated, not less real.

Task 26 should begin a new decomposition rather than another hardening pass. Its root is not "Spin" but "base plus varying metric-oriented fibres". It should first refine the deliberately weak placeholder `FiberFamilyInput` into the minimum mathematically sufficient rank-three real fibre data, distinguish intrinsic fibre structure from topology and local triviality, and determine which additional dependencies are required before a frame bundle can even be stated. Only after that varying-fibre layer is reconstructed can a later task ask whether the one-fibre $\mathcal{L} \to G_{\mathrm{vis}}$ lift extends coherently over the base. The Conservation of Difficulty has therefore moved from the internal double cover to the geometry that must carry it.

Keywords: lifted frame torsor; frame torsor; equivariant projection; central $\mathbb{Z}_2$ kernel; unit quaternions; $SU(2)$; $SO(3)$; $Spin(3)$; $U(2)$; $Spin^c(3)$; reference change; one-fibre interface; spin structure; Lean 4; formal verification; Conservation of Difficulty.

Contents

1 The last one-fibre gap

The reconstruction no longer begins from a conventional spin bundle. Its starting point was a split-complex Lorentzian structure, whose longitudinal null decomposition, visible spatial action, central residual structure, quaternionic core, and covering behavior were exposed in successive formal reductions. By Part XXIII the project possessed an exact two-sheeted group covering

$$p : \mathcal{L} \longrightarrow G_{\mathrm{vis}}, \qquad \ker p = K_{\pm} \cong \{\pm 1\}, \tag{1}$$

and had reconstructed that cover from local sections and sign-valued transition data before identifying it with standard covering-space language [8].

Part XXIV then deliberately refused to jump from (1) to a spin-structure definition. It first reconstructed the object on which the visible group acts. From the inherited three-dimensional positive-definite carrier E and its orientation datum it defined

$$\mathcal{F}^{+}(E) = \{\text{positively oriented orthonormal frames of } E\}, \tag{2}$$

without using G_{vis} or a reference frame in the definition. It proved that G_{vis} acts freely and transitively on $\mathcal{F}^{+}(E)$. A reference frame F_0 entered only afterwards as a coordinate choice. The same stage constructed a lifted-frame carrier

$$\widetilde{\mathcal{F}}_{F_0} = \{(u, F) \in \mathcal{L} \times \mathcal{F}^{+}(E) \mid p(u) \cdot F_0 = F\}, \tag{3}$$

with exactly two lifted representatives over every ordinary frame [9, 1].

The remaining defect was extremely narrow. The full internal group was known, the lifted carrier was known, and its two-element fibres were known, but the action

$$\mathcal{L} \curvearrowright \widetilde{\mathcal{F}}_{F_0} \tag{4}$$

had not been formalized. Task 24 had instead proved that the induced $\mathcal{L}$-action on the *ordinary* frame carrier is transitive but not free, with stabilizer $K_{\pm}$. That statement says nothing negative about an action on (3); the action upstairs simply did not yet exist in the formal source.

Methodological principle 1 (Do not transfer a stabilizer across different carriers)**.** A nontrivial stabilizer of the projected action on the ordinary frame carrier does not imply a nontrivial stabilizer on a lifted carrier that remembers the kernel sheet. The actions must be constructed and audited on their own carriers.

Task 25 is therefore not a new geometric branch. It is a hardening layer whose central question is exact:

> Does $\widetilde{\mathcal{F}}_{F_0}$ carry the natural free and transitive $\mathcal{L}$-action compatible with the already certified projection $p : \mathcal{L} \to G_{\mathrm{vis}}$, and is the remaining reference ambiguity exactly the same kernel $\mathbb{Z}_2$?

The answer is yes. The importance of that answer is not that it makes spin geometry appear by terminology. It closes the one-fibre algebra so completely that the missing global geometry can finally be stated without ambiguity.

2 Formal source, scope, and provenance

The supplied Task-25 archive has SHA-256

`94c70febd8e9565a862447379ecc7444c81598036ffaad99035dd5a86143d4dc`.

The pinned environment is Lean 4.28.0 with Mathlib revision

`8f9d9cff6bd728b17a24e163c9402775d9e6a365`.

No Task-24 source file is modified. The new dependency chain is

`LiftedFrameAction` → `LiftedFrameEquivariance` → `ReferenceTransferHardening` → `OneFiberInterface` → `Task25`.

There are five new modules and 978 Lean lines.

Check	Result	Detail
Focused Task-25 endpoint build	PASS	8218 jobs
Whole-project build	PASS	8289 jobs, 0 errors
Task-25 warnings	0	114 warnings are inherited from older unchanged layers
New source	5 modules	978 Lean lines
Exposed endpoints	34	all audited in `Task25.lean`
Axiom baseline	exact	`propext`, `Classical.choice`, `Quot.sound`
`sorry`/`admit`	0/0	no proof holes
Prohibited proof escapes	0	no custom `axiom`, `implemented_by`, `native_decide`, `bv_decide`, `unsafe`, or `partial`
Inherited Task-24 source	unchanged	bytewise preservation reported by the supplied audit

Proof boundary 1 (Verification boundary)**.** *The Task-25 source, audit, endpoint module, pinned toolchain and manifest, source counts, archive hash, and theorem statements were inspected for this paper. The reported 8,000-plus-job Lean builds and their axiom output are taken from the supplied Aristotle artifact; the paper-generation environment does not contain the project toolchain and did not independently rerun those builds.*

The scope firewall is unusually strict. The Task-25 source introduces no nontrivial base space, no fibre family, no tangent or vector bundle, no principal bundle, no spin structure, no topology, no characteristic class, no connection, and no physical interpretation. This negative information is mathematically central: the formal result is a one-fibre theorem, not a disguised bundle theorem.

3 The inherited one-fibre diagram

Before the new action is introduced, the inherited structure can be summarized as

$$\begin{array}{ccc} \widetilde{\mathcal{F}}_{F_0} & \xrightarrow{\pi} & \mathcal{F}^+(E) \\ \cup & & \cup \\ (u,F) & \longmapsto & F, \end{array} \tag{5}$$

with compatibility condition $p(u) \cdot F_0 = F$. The ordinary frame carrier is already a free and transitive G_{vis}-set:

$$\forall F, G \in \mathcal{F}^+(E), \qquad \exists!\, g \in G_{\mathrm{vis}} :\ g \cdot F = G. \tag{6}$$

The fibre of π over each ordinary frame contains exactly two elements, and the inherited sign action of $K_\pm$ is free on those fibres.

The internal group also acts on ordinary frames through p:

$$a \star F := p(a) \cdot F. \tag{7}$$

That action is transitive but has stabilizer $K_\pm$. This is exactly what one expects from forgetting the sheet of a double cover. Task 25 asks whether retaining that sheet restores freeness upstairs.

4 Direct construction of the full internal action upstairs

For $a \in \mathcal{L}$ and $x = (u, F) \in \widetilde{\mathcal{F}}_{F_0}$, Task 25 defines

$$a \cdot x := (au,\, p(a) \cdot F). \tag{8}$$

This is not transported through the pre-existing carrier equivalence $\widetilde{\mathcal{F}}_{F_0} \simeq \mathcal{L}$. It is constructed directly on the compatible pair. The defining condition closes because

$$\begin{aligned} p(au) \cdot F_0 &= p(a)p(u) \cdot F_0 && (9)\\ &= p(a) \cdot (p(u) \cdot F_0) && (10)\\ &= p(a) \cdot F. && (11) \end{aligned}$$

The identity and multiplication laws are then proved and packaged as a genuine `MulAction`.

Theorem 1 (Free lifted-frame action)**.** *For every $F_0 \in \mathcal{F}^+(E)$, $a \in \mathcal{L}$, and $x \in \widetilde{\mathcal{F}}_{F_0}$,*

$$a \cdot x = x \quad \Longrightarrow \quad a = 1. \tag{12}$$

The proof is elementary but decisive. Equality upstairs includes equality of the internal components, hence $au = u$; cancellation gives $a = 1$. The sign that disappears downstairs cannot disappear upstairs because it multiplies the remembered internal representative.

Theorem 2 (Transitivity and unique transporter)**.** *For any $x, y \in \widetilde{\mathcal{F}}_{F_0}$ there exists exactly one $a \in \mathcal{L}$ with $a \cdot x = y$. The explicit transporter is*

$$a = y_{\mathrm{elt}} x_{\mathrm{elt}}^{-1}. \tag{13}$$

Thus

$$\forall x, y \in \widetilde{\mathcal{F}}_{F_0}, \qquad \exists!\, a \in \mathcal{L} :\; a \cdot x = y. \tag{14}$$

At the elementary group-action level, $\widetilde{\mathcal{F}}_{F_0}$ is therefore a torsor for $\mathcal{L}$.

Status 1 (The Task-24 review gap is closed)**.** The ordinary frame carrier is a free transitive G_{vis}-space, while the lifted frame carrier is a free transitive $\mathcal{L}$-space. The earlier non-freeness of the projected $\mathcal{L}$-action on ordinary frames remains true and is now correctly located: it is a property of the projected carrier, not of the lifted one.

5 Equivariance and the exact role of the kernel

The new action is tied to the old visible action by the exact identity

$$\pi(a \cdot x) = p(a) \cdot \pi(x). \tag{15}$$

Equivalently, the square

$$\begin{tikzcd} \mathcal{L} \times \widetilde{\mathcal{F}}_{F_0} \arrow[r, "\text{action}"] \arrow[d, "p \times \pi"'] & \widetilde{\mathcal{F}}_{F_0} \arrow[d, "\pi"] \\ G_{\mathrm{vis}} \times \mathcal{F}^+(E) \arrow[r, "\text{action}"'] & \mathcal{F}^+(E) \end{tikzcd}$$

commutes formally.

The kernel is then identified at the action level, not merely at the group level. For $\varepsilon \in K_\pm$,

$$\varepsilon \cdot x = \mathrm{signSmul}(\varepsilon, x), \tag{16}$$

where the right-hand side is the vertical sign action inherited from Part XXIV. Therefore

$$\pi(\varepsilon \cdot x) = \pi(x), \tag{17}$$
$$\varepsilon \cdot x = x \Longrightarrow \varepsilon = 1. \tag{18}$$

The same two-element subgroup is simultaneously

- the kernel of the group projection p;
- the stabilizer of the projected $\mathcal{L}$-action on ordinary frames;
- the free vertical deck-like action on each lifted-frame fibre;
- the exact residual ambiguity of lifted reference change.

These roles are now linked by formal theorems rather than by analogy.

6 The carrier equivalence becomes an equivariant equivalence

Part XXIV had already established, for fixed F_0, an equivalence of carriers

$$\widetilde{\mathcal{F}}_{F_0} \simeq \mathcal{L}. \tag{19}$$

A bare bijection is weaker than a torsor identification. Task 25 proves that (19) intertwines the new left action upstairs with left multiplication on $\mathcal{L}$:

$$\Phi(a \cdot x) = a\,\Phi(x). \tag{20}$$

Its inverse satisfies the corresponding identity.

The source also contains a negative control: another carrier equivalence can be constructed that is not equivariant. This establishes that equivariance is genuine structure, not a property of every bijection between the two carriers.

Methodological principle 2 (Equivalence of carriers is not equivalence of actions)**.** The statement $X \simeq G$ does not make X a G-torsor. The action, its freeness and transitivity, and compatibility of the equivalence with multiplication must be proved separately.

7 Reference change: unique downstairs, exactly two-valued upstairs

Let $F_0, F_1 \in \mathcal{F}^+(E)$. The visible torsor theorem gives exactly one

$$g_{01} \in G_{\mathrm{vis}}, \qquad g_{01} \cdot F_0 = F_1. \tag{21}$$

Task 25 considers the fibre

$$\mathrm{Lifts}(g_{01}) := \{u \in \mathcal{L} \mid p(u) = g_{01}\}. \tag{22}$$

It proves

$$\#\,\mathrm{Lifts}(g_{01}) = 2. \tag{23}$$

Moreover, for any two u, v in this fibre there exists exactly one $\varepsilon \in K_\pm$ with

$$v = \varepsilon u. \tag{24}$$

Thus the possible lifted reference changes form one free transitive $K_\pm$-orbit.

Each chosen lift u induces the inherited transfer map

$$T_u : \widetilde{\mathcal{F}}_{F_0} \longrightarrow \widetilde{\mathcal{F}}_{F_1}. \tag{25}$$

The multiplication convention matters. The transfer acts by right multiplication with u^{-1} on the internal representative, whereas the newly defined $\mathcal{L}$-action acts on the left. The two therefore commute:

$$T_u(a \cdot x) = a \cdot T_u(x). \tag{26}$$

No conjugation correction is needed.

If u, v are the two possible lifts of the same visible reference change, the corresponding transfers satisfy

$$\exists!\, \varepsilon \in K_\pm : \quad T_v(x) = \varepsilon \cdot T_u(x) \quad \text{for all } x. \tag{27}$$

The converse sign-shift statement is also proved.

Status 2 (Reference-change classification)**.** Downstairs reference change is unique. Upstairs it is not unique and should not be made unique by an arbitrary normalization. The entire ambiguity is the already reconstructed central $\mathbb{Z}_2$ kernel.

A precision boundary remains worth recording. Task 25 classifies exactly the transfer equivalences arising from the two lifts of the unique downstairs reference change. It does not separately prove that *every imaginable* interface automorphism between two presentations must arise in this way. Nothing in the present paper requires that stronger classification.

8 Free upstairs, non-free after projection

The project can now display the two actions side by side:

Carrier	Acting group	Stabilizer / transport
$\mathcal{F}^+(E)$	G_{vis}	free and transitive; unique g between frames
$\mathcal{F}^+(E)$	$\mathcal{L}$ through p	transitive; stabilizer exactly $K_\pm$
$\widetilde{\mathcal{F}}_{F_0}$	$K_\pm$ vertically on a fixed fibre	free and transitive on the two-point fibre
$\widetilde{\mathcal{F}}_{F_0}$	full $\mathcal{L}$	free and transitive; unique a between lifted frames

This table explains the role of the sign layer without invoking physical spin. Forgetting the lifted representative identifies u and $-u$; retaining it makes the internal group action faithful at the point level. The passage from upstairs to downstairs is therefore not merely "two copies" of the same object. It is an equivariant quotient in which the kernel becomes invisible.

9 Freezing the one-fibre interface

Task 25 packages the completed structure as `OneFiberLiftInterface`. Conceptually it contains

$$(\widetilde{\mathcal{F}},\ \mathcal{L} \curvearrowright \widetilde{\mathcal{F}},\ \pi : \widetilde{\mathcal{F}} \to \mathcal{F}^+(E),\ G_{\text{vis}} \curvearrowright \mathcal{F}^+(E),\ p : \mathcal{L} \to G_{\text{vis}}), \tag{28}$$

together with:

- a nonempty lifted carrier;

- the action laws;
- free and transitive internal action;
- projection equivariance;
- surjectivity of the projection;
- the theorem that two lifted points project to the same ordinary frame exactly when they differ by an element of $K_{\pm}$;
- the inherited free/transitive visible action downstairs.

Crucially, the temporary reference frame is not a field of the interface. Neither is a chosen lift of a reference change. They live in separate presentation records. For any two temporary reference frames, Task 25 proves existence of an interface equivalence, and two equivalences produced by the two lift choices differ by exactly one kernel sign.

Methodological caveat 1 (What "reference independent" means here)**.** *The formal result is reference independence up to an explicitly classified $\mathbb{Z}_2$ family of comparison maps. No quotient of all reference choices is constructed, and no unique preferred comparison map is claimed.*

This is the correct stopping point for the one-fibre problem. The interface records exactly the structure that future varying-fibre work may consume without reopening the internal reconstruction.

10 Engineering provenance and failed builds

Three failed builds are preserved in the Task-25 audit. None changed a theorem statement or exposed a false mathematical expectation.

#	Failure	Diagnosis	Repair
1	Rewrite failure, deterministic timeout, then unknown constant in the equivariance module	An existential goal had not beta-reduced into a rewrite-friendly shape; an attempted inverse-equivariance proof forced expensive unfolding of a noncomputable projection	Added explicit goal shapes and proved inverse equivariance via injectivity and `Equiv.apply_symm_apply` instead of subtype extensionality
2	Unknown ambient identifier, invalid substitution, unresolved set membership and action orientation in reference hardening	Proof script depended on names not opened in the new namespace and used an unsuitable witness/rewrite direction	Reproved kernel involution abstractly from the two-element kernel theorem; corrected the witness; unfolded pair-set membership explicitly; used an explicit equality chain
3	Failed action-instance synthesis through the abstract interface carrier	The interface field `Carrier` is not definitionally reducible to the concrete `LiftedFrame` type for instance search	Stated the ambiguity theorem on the concrete transfer maps and added an `rfl` bridge to the interface equivalence

These failures are useful engineering provenance. In particular, the third failure illustrates a recurring formal distinction: an abstract interface can preserve mathematical structure while hiding enough definitional information to prevent automatic instance synthesis. The repair retained the abstract interface rather than weakening it for proof convenience.

11 Comparison with established mathematics

Only now, after the intrinsic one-fibre structure is frozen, is it useful to compare the result with conventional geometry.

11.1 The group double cover

Classically, the unit quaternions form a group isomorphic to $SU(2)$, and conjugation on imaginary quaternions gives the standard two-to-one homomorphism

$$SU(2) \longrightarrow SO(3) \tag{29}$$

with kernel $\{\pm I\}$ [3, 4, 6]. In standard notation this is the three-dimensional spin covering

$$Spin(3) \cong SU(2) \longrightarrow SO(3). \tag{30}$$

The project has reconstructed the group-theoretic substrate of (29) internally and then compared it externally:

- $\mathcal{L}$ is formally equivalent to the unit quaternions and to $SU(2)$;
- G_{vis} is formally equivalent to $SO(3)$;
- the intrinsic spatial action is formally identified with quaternionic conjugation;
- $\ker p = \{\pm 1\}$;
- the projection is formally a two-sheeted covering map after the local reconstruction.

The name $Spin(3)$ itself is not packaged as a formal Lean equivalence in the project. Its use here is therefore an established-mathematics comparison, not an additional formal theorem.

11.2 The ordinary frame torsor

For a fixed oriented Euclidean three-space, the set of oriented orthonormal frames is a torsor for $SO(3)$: choosing a reference frame identifies the torsor with the group, but the torsor itself does not contain a preferred origin [5]. This is now exactly the role of $\mathcal{F}^+(E)$:

$$\forall F, G \in \mathcal{F}^+(E), \qquad \exists!\, g \in G_{\text{vis}} : \; g \cdot F = G. \tag{31}$$

A reference frame supplies $G_{\text{vis}} \simeq \mathcal{F}^+(E)$ only after the free/transitive theorem is established.

11.3 The lifted frame torsor

At the one-fibre level, Task 25 now supplies the corresponding lifted torsor:

$$\forall \widetilde{F}, \widetilde{G} \in \widetilde{\mathcal{F}}_{F_0}, \qquad \exists!\, u \in \mathcal{L} : \; u \cdot \widetilde{F} = \widetilde{G}, \tag{32}$$

with an equivariant two-to-one projection to ordinary frames. Under the established comparison $\mathcal{L} \cong Spin(3)$ and $G_{\text{vis}} \cong SO(3)$, this is pointwise the standard algebraic pattern expected above an oriented orthonormal frame fibre.

That phrase "pointwise" is essential. Classical spin geometry does not define a spin structure as a pair of torsors over one point. It defines a lift of an oriented orthonormal frame *bundle* over a base manifold or, more generally, of an oriented metric vector bundle [6, 7].

11.4 The extended central group

The earlier central-extension branch established

$$\mathcal{L}_Z \simeq \mathbb{R}_{>0} \times U(2) \tag{33}$$

at the abstract group level, with the normalized carrier equivalent to $U(2)$. Classically,

$$Spin^c(3) \cong (Spin(3) \times U(1))/\mathbb{Z}_2 \cong U(2). \tag{34}$$

Thus the project has also reached the conventional $Spin^c(3)$ group shape at the group-comparison level. Again, no $Spin^c$ structure on a bundle or manifold has been constructed. Group-level coincidence and geometric structure must remain separate.

12 What has already been reconstructed, and what has not

Established mathematical object or relation	Project status	Boundary
Split-complex/null-coordinate longitudinal Lorentz structure	FORMAL	reconstructed at the start of the chain
Three-dimensional metric spatial carrier with orientation datum	FORMAL	inherited before frame reconstruction
Unit-quaternion internal core	FORMAL	exact carrier/group equivalence
$SU(2)$ model of the internal core	FORMAL	exact group equivalence
Visible rotation group $SO(3)$	FORMAL	exact group equivalence
Quaternionic rotation action	FORMAL	identified with intrinsic projection/action
Kernel $\{\pm 1\}$	FORMAL	exact kernel theorem
Two-sheeted covering $\mathcal{L} \to G_{\text{vis}}$	FORMAL	reconstructed locally then identified as a covering map
Local sections and $\mathbb{Z}_2$ transition signs	FORMAL	exact overlap and reconstruction laws
Oriented orthonormal frame torsor for G_{vis}	FORMAL at set/group-action level	no topology on $\mathcal{F}^+(E)$
Lifted frame torsor for $\mathcal{L}$	FORMAL at set/group-action level	new in Task 25; no topology on $\widetilde{\mathcal{F}}_{F_0}$
Equivariant two-sheeted projection of frame torsors	FORMAL at set/group-action level	$\pi(a \cdot x) = p(a) \cdot \pi(x)$
Reference changes upstairs classified by $\mathbb{Z}_2$	FORMAL	exactly two lift choices, unique kernel relation
$Spin(3) \cong SU(2)$	ESTABLISHED COMPARISON	named $Spin(3)$ object not formalized in the project
Normalized extended group $\cong U(2)$	FORMAL	abstract group equivalence
$Spin^c(3) \cong U(2)$	ESTABLISHED COMPARISON	named $Spin^c$ object not formalized
Nontrivial base space B	ABSENT	deliberately not introduced
Rank-three real vector bundle over B	ABSENT	not even defined yet

Established mathematical object or relation	Project status	Boundary
Oriented orthonormal frame bundle over B	ABSENT	requires varying fibres and local triviality
Principal $SO(3)$ bundle structure	ABSENT	no total bundle/topology
Principal $Spin(3)$ lift	ABSENT	the defining global spin-structure datum is missing
Second Stiefel–Whitney obstruction w_2	ABSENT	no bundle on which it could yet be evaluated
Spin structure on a manifold	ABSENT	not claimed
Physical spin state or dynamics	ABSENT	outside the formal content

This table captures the central tension of Part XXV. Many of the algebraic and pointwise ingredients recognizable from spin geometry have indeed been reconstructed. The missing object is no longer vague. It is the coherent variation of these fibres over a base.

13 Why this is still not a spin structure

For an oriented Riemannian rank-n vector bundle $E \to M$, standard spin geometry begins with the principal oriented orthonormal frame bundle

$$P_{SO}(E) \longrightarrow M \tag{35}$$

and the double covering homomorphism

$$\rho : Spin(n) \longrightarrow SO(n). \tag{36}$$

A spin structure consists, in one standard formulation, of a principal $Spin(n)$ bundle

$$P_{Spin}(E) \longrightarrow M \tag{37}$$

together with an equivariant two-to-one map

$$\lambda : P_{Spin}(E) \longrightarrow P_{SO}(E) \tag{38}$$

satisfying

$$\lambda(q \cdot s) = \lambda(q) \cdot \rho(s). \tag{39}$$

See, for example, Lawson–Michelsohn and the standard theory of characteristic classes [6, 7].

Task 25 has reconstructed the *fibrewise algebraic shadow* of (38) for one fixed three-dimensional metric-oriented carrier. It has not reconstructed the following dependencies:

1. a nontrivial base M or B;
2. a rank-three real vector space E_b at each base point;
3. coherent vector-space, metric, and orientation structure across the family;
4. a total ordinary frame space over the base;
5. a topology or smooth structure on that total space;
6. local trivializations and their transition functions;
7. a principal $SO(3)$ bundle action on the total frame space;

8. a corresponding total lifted space;

9. compatible lifted transition functions or a principal $Spin(3)$ action over the base;

10. the global equivariant covering map of total spaces.

Without these, there is no object to which the usual existence obstruction w_2 could even be attached. The absence of a w_2 calculation is therefore not a missing final trick. It reflects an earlier dependency boundary: the relevant bundle has not yet been constructed.

Methodological principle 3 (Pointwise spin data are not a spin structure)**.** A double-cover group, an ordinary frame torsor, a lifted torsor, and an equivariant two-to-one map over one point reproduce the fibrewise algebra of a spin structure. A spin structure additionally requires those fibres to vary coherently over a base as principal bundles. The latter cannot be inferred from the former without new geometric input or reconstruction.

14 The exact frontier exported to Task XXVI

Task 25 deliberately introduces a neutral record `FiberFamilyInput` with a base type, a fibre assignment, a fibrewise two-argument real-valued datum called a metric, and an orientation predicate. It is explicitly uninstantiated. This record should not be mistaken for a finished mathematical model of a rank-three oriented Euclidean vector bundle. In particular, it does not yet assert real vector-space structure, dimension three, bilinearity, symmetry, positive definiteness, or any compatibility across base points.

That weakness is useful because it prevents Task 26 from inheriting hidden structure. The next task should begin by asking which of these ingredients are definitions, which are additional data, and which must be proved.

A focused Task 26 should therefore begin from the root

$$\boxed{\text{base } B \ + \ \text{varying metric-oriented rank-three fibres}}, \tag{40}$$

not from a prepackaged principal bundle or spin structure. Its first obligations should be only:

1. refine the minimum fibre-family input to genuine fibrewise real three-dimensional inner-product spaces with orientation;

2. construct the fibrewise ordinary-frame carrier $b \mapsto \mathcal{F}^+(E)_b$ without choosing local frames;

3. construct the total set $\bigsqcup_{b\in B} \mathcal{F}^+(E)_b$ and its projection to B;

4. classify exactly what extra topology/local-triviality data are needed before this total carrier can be called a frame bundle;

5. if local trivializations are introduced, derive rather than assume the corresponding G_{vis}-valued transition maps and their overlap law.

It should stop before solving the lifted-bundle problem unless the ordinary varying-frame layer closes cheaply. The one-fibre interface of Task 25 is then the local algebraic model that each fibre must reproduce.

This ordering preserves the central methodology of the project: reconstruct the object first, identify its dependencies, and only then compare with the conventional package.

15 Conservation of Difficulty

The sequence from the first spin-like sign behavior to Part XXV can now be read as a migration of the unresolved datum rather than its disappearance.

Boundary	Where the difficulty moved
Internal $2\pi/4\pi$ sign behavior	From a suggestive periodicity to a global representative/continuity obstruction
Local exact representatives	From existence to overlap compatibility and global reconstruction
Certified double cover	From the group homomorphism to the geometric object acted upon
Intrinsic ordinary frame carrier	From basis/reference choice to a free-transitive visible action
Two-valued lifted frames	From fibre cardinality to the missing full internal action
Task 25 free-transitive lifted action	From one-fibre algebra to coherent variation over a base
Future varying frame family	Expected next boundary: topology, local triviality, transition data, and only later a global lifted bundle

The present stage is therefore both more classical and more constrained than the earlier spin-like comparisons. The core group, visible group, covering map, frame torsor, lifted torsor, and kernel ambiguity now line up with familiar three-dimensional spin geometry. The project has not "almost proved a spin structure" in the sense that one final theorem is missing. Rather, it has completely solved the algebraic one-fibre layer and exposed a qualitatively different global-geometric layer as the next dependency class.

16 Decision table and theorem boundary

Question	Verdict
Natural $\mathcal{L}$ action on $\widetilde{\mathcal{F}}_{F_0}$ defined?	PROVED
Action laws?	PROVED
Action free?	PROVED
Action transitive?	PROVED
Unique internal transporter between lifted frames?	PROVED
Projection to ordinary frames equivariant?	PROVED
Kernel restriction equals inherited sign action?	PROVED
Kernel signs trivial downstairs?	PROVED
Kernel signs free upstairs?	PROVED
Carrier equivalence $\widetilde{\mathcal{F}}_{F_0} \simeq \mathcal{L}$ action-equivariant?	PROVED
Unique reference change downstairs?	INHERITED / PROVED earlier
Exactly two lifted reference changes?	PROVED
Difference of lifted reference changes exactly one kernel sign?	PROVED
Lifted transfer equivariant?	PROVED
Two transfer maps differ by unique kernel sign?	PROVED

Question	Verdict
One-fibre interface independent of temporary reference presentation?	PROVED up to classified $\mathbb{Z}_2$ ambiguity
Topology on ordinary/lifted frame carriers?	NOT CONSTRUCTED
Nontrivial varying base?	NOT CONSTRUCTED
Frame bundle?	NOT CONSTRUCTED
Principal bundle?	NOT CONSTRUCTED
Spin structure?	NOT CONSTRUCTED
Physical spin degree of freedom?	NOT DERIVED

17 Conclusion

Part XXV closes a very specific historical gap and, by doing so, changes the nature of the remaining problem. The lifted-frame carrier is no longer merely a two-valued set sitting above the ordinary frame carrier. It carries the direct full action of the reconstructed internal group, that action is free and transitive, its projection is equivariant along the certified double cover, and the kernel action is exactly the vertical sign action already exposed by the local-cover analysis. The ordinary and lifted frame carriers are therefore related by a complete algebraic one-fibre diagram.

Reference dependence also becomes exact. A reference frame is still only a presentation choice. The visible reference change is unique; its internal lifts are exactly two; and the resulting equivariant transfer maps differ by exactly one kernel sign. The surviving $\mathbb{Z}_2$ ambiguity is not an error to remove but the same structural kernel that has persisted through the entire reconstruction.

The late classical comparison is now correspondingly strong. The internal group has the unit-quaternion/$SU(2)$ form, the visible group the $SO(3)$ form, the projection the standard quaternionic two-fold pattern, ordinary oriented frames form the expected visible torsor, and lifted frames form the corresponding internal torsor. At the group level, these are the ingredients associated with $Spin(3) \to SO(3)$; the extended normalized group also has the classical $Spin^c(3) \cong U(2)$ shape. But a spin group is not a spin structure, and a lifted torsor over one point is not a lifted frame bundle over a manifold.

The missing dependency is now sharply localized. Task 26 must leave the completed one-fibre algebra untouched and begin a fresh reconstruction of variation over a base. Until a genuine family of rank-three metric-oriented fibres, its total frame space, topology/local triviality, and transition structure have been constructed, there is still no frame bundle to lift and hence no spin structure to claim. The Conservation of Difficulty has moved from internal algebra to global geometry.

A Task-25 principal theorem map

The 34 exposed Task-25 endpoints are organized into four mathematical blocks:

1. **Full action upstairs:** `liftSmul`, identity, multiplication, freeness, transitivity, unique transporter, and the explicit free-vs-projected comparison.
2. **Equivariance and kernel:** projection equivariance, commuting square, exact restriction to `KerSign`, trivial action downstairs, free action upstairs, and free/transitive kernel fibres.

3. **Reference-change hardening:** equivariance of the inherited carrier equivalence, two lifts of the unique reference change, unique kernel relation, equivariant transfers, and unique-sign comparison of transfer maps.

4. **Interface freeze and controls:** `OneFiberLiftInterface`, reference-independent interface equivalence, classified ambiguity, one-fibre-only theorem, and negative controls distinguishing carrier equivalence from equivariant equivalence and downstairs uniqueness from upstairs uniqueness.

Every exposed endpoint reports exactly the inherited axiom baseline

$$[\texttt{propext},\ \texttt{Classical.choice},\ \texttt{Quot.sound}]. \tag{41}$$

B Exact unresolved obligations after Task XXV

The Task-25 audit lists four unresolved classes:

1. no topology on $\mathcal{F}^+(E)$ or $\widetilde{\mathcal{F}}_{F_0}$, hence no topological torsor or local-triviality theorem at this level;

2. no quotient of all reference presentations into a single presentation-free carrier, because equivalence up to classified sign is sufficient for the present interface;

3. no re-analysis of the size or internal structure of $\mathcal{L}$ beyond the inherited group theorems needed here;

4. all varying-base structures remain absent by design.

These are not hidden proof holes. They are the explicit mathematical boundary of the completed task.

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